

EXTENDED READING AND STATIC VERIFICATION

Better advice, weaker learning?

AI, Human Cognition and Knowledge Collapse

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NBER Working Paper 34910, February 2026

github.com/gsaco/ai-04-acemoglu

Static derivations and an original assumption test.

Dynamic and welfare results are reported from the paper, not reproduced.

The bridge: private decisions create public information

The information that saves effort today can weaken learning tomorrow.

- A person needs general understanding and context-specific information.
- Human effort improves their own information and leaves a public trace.
- Personalized AI replaces part of the private reason to learn.
- That response can reduce the public knowledge inherited by later cohorts.

Introduction; §2, pp. 5–6: the contribution combines individual information acquisition with collective learning, rather than treating collaboration as a single isolated task.

Production: complementarity is an assumption

Let $b_G, b_I \in \{0, 1\}$ indicate prediction errors within a unit band.

$$f(b_G, b_I) = f(0, 0) + \Delta_G b_G + \Delta_I b_I + \Delta_X b_G b_I.$$

$$\Delta_G = f(1, 0) - f(0, 0),$$

$$\Delta_I = f(0, 1) - f(0, 0),$$

$$\Delta_X = f(1, 1) - f(1, 0) - f(0, 1) + f(0, 0).$$

Monotone f ; $\Delta_G + \Delta_I + \Delta_X = 1$.

Assumption 1: $\Delta_I = 0$, $\Delta_X > 0$. General knowledge is essential for context-specific information to pay off.

§3.1, p. 8. Leontief is one example ($\Delta_G = \Delta_I = 0$, $\Delta_X = 1$), not the entire model.

Normal information: precisions add

The common state drifts; the individual state is fresh each period:

$$\theta_{t+1} = \theta_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \Sigma^2), \quad \theta_{i,t} \sim N(0, \sigma^2).$$

Independent human and AI signals about the individual state give

$$Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

Public information: $X = \text{Var}(\theta_t \mid \mathcal{I}_t)^{-1}$. It is inherited when effort is chosen.

Posterior means maximize the probability of lying within the symmetric unit-tolerance band. The two success probabilities are $G(X)$ and $G(Y)$.

§§3.1–3.3, pp. 7–12, eqs. (4)–(5). Gaussian, conditionally independent information.

The static objective and its unique optimum

Dropping $f(0,0)$, and using $\Delta_I = 0$:

$$U = G(X)\Delta_G + G(X)G(Y)\Delta_X - e^\alpha/\alpha.$$

$$\Delta_X G(X) \lambda_I g(Y) = e^{\alpha-1}.$$

For $e > 0$,

$$U_{ee} = \Delta_X G(X) \lambda_I^2 g'(Y) - (\alpha - 1)e^{\alpha-2} < 0.$$

At $X > 0$, marginal benefit at zero is positive, while marginal cost ultimately dominates: a unique, finite, interior optimum exists.

§3.5, p. 14. Conditions: $\alpha > 1$, $\lambda_I > 0$, $\Delta_X > 0$,

$\sigma^{-2} > 0$, finite $\tau_A \geq 0$. At $X = 0$, the optimum is the boundary $e = 0$.

Deriving the two signs by hand

$$G(z) = 2\Phi(\sqrt{z}) - 1, \quad g(z) = \frac{e^{-z/2}}{\sqrt{2\pi z}}.$$

Log-differentiation makes the curvature transparent:

$$\frac{g'(z)}{g(z)} = -\frac{1}{2} - \frac{1}{2z} < 0.$$

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0,$$

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

Interpretation: X increases the value of private precision; τ_A reduces its marginal value by already supplying it.

Observation 1, p. 13. Strict signs on $X, Y > 0$ with $\Delta_X, \lambda_I > 0$;

short-lived atomistic agents do not internalize their public learning contribution.

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From payoff cross-partials to effort responses

Differentiate the interior FOC $U_e(e^*, X, \tau_A) = 0$:

$$e_X^* = -\frac{U_{eX}}{U_{ee}} > 0, \quad e_{\tau_A}^* = -\frac{U_{e\tau_A}}{U_{ee}} < 0.$$

Strict payoff concavity turns the cross-partial signs into effort responses.

- These derivatives hold at fixed inherited X or fixed τ_A .
- Aggregation capacity I is absent from the current private FOC.
- I matters through the public information later cohorts inherit.

Observation 2, p. 14; own implicit-function derivation.

Do not confuse the technology parameter λ_I with island size I .

The Bayesian bridge, read without solving dynamics

The paper's public-information recursion is

$$V_{t+1} = \frac{1}{X_t + \lambda_G l e_t} + \Sigma^2, \quad X_{t+1} = \left[\frac{1}{X_t + \lambda_G l e_t} + \Sigma^2 \right]^{-1}.$$

- Pooling human signals adds **precision**: $\lambda_G l e_t$.
- State innovation adds **variance**: Σ^2 .
- The static best response supplies $e_t = e(X_t, \tau_A)$.

§3.2, eq. (3), p. 11; Proposition 1, p. 15.

Reported model equation only: no recursion simulation or steady-state reconstruction.

What the dynamic results claim

Assumption 1; $I, \lambda_I, \lambda_G, \Sigma^2, \sigma^2 > 0$; finite $\tau_A \geq 0$.

Cost regime	Reported result
$\alpha - 1 > 1/4$	Zero is unstable; one positive stable steady state attracts every $X_1 > 0$. Zero remains a fixed point.
$0 < \alpha - 1 < 1/4$	Below τ_A^c , zero and a high state are stable, separated by $\bar{X}_m$. Above τ_A^c , only zero remains.

In the second regime, $X_1 > \bar{X}_m$ selects the high state; $X_1 < \bar{X}_m$ selects collapse. At $X_1 = \bar{X}_m$, the state stays there.

Boundary discipline: these claims exclude $\alpha - 1 = 1/4$. We use strict inequalities around τ_A^c ; the paper's treatment of equality is inconsistent.

Propositions 3 and 5, pp. 18, 20; see extra/analysis.md for the threshold caveat.

Welfare: which quantity is being differentiated?

Static optimized utility $v(X, \tau_A) = \max_e U(e; X, \tau_A)$:

$$\left. \frac{\partial v}{\partial \tau_A} \right|_X = G(X) \Delta_X g(Y) > 0.$$

High-steady-state welfare, as reported in the paper:

$$\frac{d\bar{U}^+}{d\tau_A} = \underbrace{G(\bar{X}_h) \Delta_X g(\bar{Y}_h)}_{\text{direct gain}} + \underbrace{g(\bar{X}_h) [\Delta_G + G(\bar{Y}_h) \Delta_X]}_{\text{loss of public knowledge}} \frac{d\bar{X}_h}{d\tau_A}.$$

Private improvement at fixed knowledge does not imply a social long-run improvement.

Static envelope calculation: own verification. Long-run decomposition: §4.3, p. 25.

Welfare is output minus effort cost, relative to $f(0, 0)$; dynamic results are read-only.

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Welfare theorems: the qualifications are the result

Common conditions: baseline model and Assumption 2, $\sigma^{-2} \geq \sqrt{2} - 1$; other parameters held fixed.

Proposition 10: if $\alpha - 1 > 1/4$, a finite $\tau_A^* \geq 0$ maximizes $\bar{U}^+$. Welfare increases below it, decreases above it, and tends to zero as $\tau_A \rightarrow \infty$. Realized high-state welfare requires $X_1 > 0$.

Proposition 11: if $0 < \alpha - 1 < 1/4$ and the positive branch exists, $0 \leq \tau_A^* < \tau_A^c$. Welfare on that branch increases then decreases; above τ_A^c the collapse outcome has zero normalized welfare. Realized welfare also depends on the initial basin.

Not necessarily interior: $\tau_A^* = 0$ is explicitly allowed. Without Assumption 2, the paper discusses failure of single-peakedness.

§4.1, pp. 23–24; Assumption 2 and footnote 6, p. 25; Propositions 10–11, p. 26.

When no positive branch exists even at zero AI precision, do not assert a positive-branch optimum.

Section 5 changes information production

Extension	What changes and what is claimed
5.1 Aggregation	$I(\tau_A) = I_0 + e^{\eta\tau_A}$. For $\eta < 1/[2(\alpha - 1)]$, knowledge still tends to zero at very high AI precision.
5.2 Synthetic data	Adds $0 < \tau_{\text{syn}} < \infty$ public precision each period. Least and greatest steady states are positive; effort and knowledge fall with τ_A .
5.3 Effort bundling	Public precision scales with e^β . For $\beta > 0$, the stability comparison becomes $\alpha - 1 \gtrless \beta/4$. $\beta = 0$ is exceptional.

Never relaxed: $\Delta_I = 0$. Section 5.3 changes the public-learning technology; it does not give context-specific knowledge standalone output value.

§5, Propositions 14–16, pp. 29–32. The formal §5.3 extension has one effort choice, not a separately optimized allocation between two effort types.

My extension: give context-specific knowledge some value

Allow $\Delta_I > 0$, retain $\Delta_X > 0$, and preserve normalization:

$$U = \Delta_G G(X) + [\Delta_I + \Delta_X G(X)]G(Y) - e^\alpha / \alpha.$$

$$U_{eX} = \Delta_X \lambda_I g(X)g(Y) > 0,$$

$$U_{e\tau_A} = [\Delta_I + \Delta_X G(X)]\lambda_I g'(Y) < 0.$$

At the zero-knowledge boundary and finite τ_A ,

$$U_e(0; 0, \tau_A) = \Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0.$$

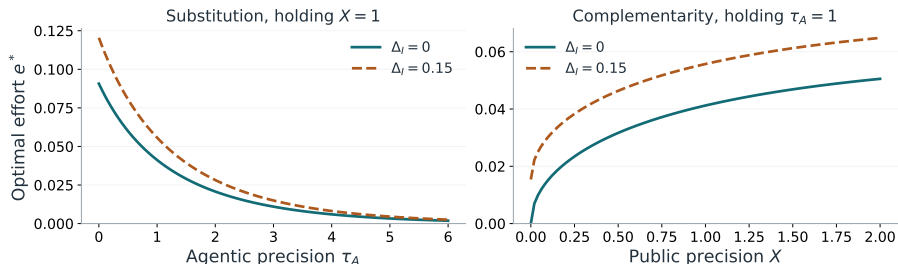
Verdict: both signs survive; zero effort at zero knowledge does not. This tests the essential-input assumption, without solving a new dynamic model.

Own static extension, not a result stated by the authors.

$\alpha > 1$, $\lambda_I > 0$, $\sigma^{-2} > 0$, $\Delta_I > 0$; choose $\Delta_G \geq 0$ to keep monotonicity.

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Reproducible evidence, not a calibration



Six symbolic identities; 444 numerical optima; FOC residuals below 2.5×10^{-15} . Implicit-function and envelope derivatives match finite differences.

At $X = 0$, $\tau_A = 1$: $e^* = 0$ under $\Delta_I = 0$ versus 0.01539 under $\Delta_I = .15$.

extra/static_checks.py; $\alpha = 2$, $\lambda_I = \sigma^{-2} = 1$, $\Delta_X = .6$, $\Delta_G = 1 - \Delta_I - \Delta_X$.

Illustrative parameters only. No dynamic paths, collapse thresholds or welfare curves fitted.

Handwritten static derivation

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$$Y = \sigma^2 + \lambda_1 e + \tau_A, \quad G(z) = 2\phi(\sqrt{z}) - 1$$
$$g(z) = G'(z) = 2\phi'(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} = \frac{\phi'(\sqrt{z})}{\sqrt{z}}$$
$$\log g(z) = -\frac{1}{2} - \log \sqrt{z} - \log(\pi/\sqrt{z})$$
$$g'(z) = -\frac{1}{2} \left(\frac{1}{z} \right) g(z) < 0$$

Con $\Delta z = 0 \rightarrow \Delta \tau > 0$

$$U = \Delta_G G(X) + \Delta_X G(X) G(Y) - \frac{e^*}{\alpha}$$
$$Y_G = \Delta \tau \quad U_G = \Delta_X G(X) \lambda_1 g(Y) - e^* = 0$$
$$U_{eX} = \Delta_X \lambda_1 g(X) g(Y) > 0$$
$$U_{eA} = \Delta_X G(X) \lambda_1 g(Y) < 0$$

Derivation checklist:

1. Differentiate G and g .
2. Derive the FOC and both signs.
3. Add $\Delta_I G(Y)$ and evaluate the boundary marginal return.

Own verdict to defend:

The static crowd-out mechanism is robust to $\Delta_I > 0$. Exact zero-knowledge collapse and zero welfare rely on additional assumptions.

Student-provided photograph: [hand/derivation.jpg](#). Baseline derivatives and FOC are handwritten; the extension and verdict are presented separately in the slides and checked in the analysis.

Limits, policy claims and questions for the oral

Aggregation: Proposition 9 raises high-state welfare with I , when that state exists. Better aggregation and better agentic advice are different changes.

Policy, read-only: Proposition 13 optimizes eventual Gaussian signal caps under long-run average welfare, $0 < \alpha - 1 < 1/4$, Assumption 2, and $X_1 > \bar{X}_m(0)$. It is not an unrestricted or discounted policy theorem.

Questions to defend:

- Why does zero effort follow at $X = 0$? Because $\Delta_I = 0$.
- Does $\Delta_I > 0$ reverse crowd-out? No, the negative cross-partial remains.
- Does the example prove AI always improves long-run welfare? No.
- Is an interior accuracy cap guaranteed? No; $\tau_A^* = 0$ is allowed.

Primary source throughout: supplied w34910.pdf, NBER WP 34910 (February 2026).

Detailed conditions, derivations, evidence map and five-minute script: [extra/analysis.md](#) and [extra/oral-notes.md](#).

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